

Prediction

Diagnostics summary

Before fitting a model:

- + Check empirical logit plots; make transformations as needed
- + Check for multicollinearity with scatterplots / correlations between quantitative explanatory variables; if needed, choose a subset of variables or combine variables

After fitting a model:

- + Check quantile residual plots
- + Check for multicollinearity with VIFs
- + Check for influential points with Cook's distance
 - + If influential points are clearly an error, can remove
 - + Otherwise, report results with and without influential points

Using logistic regression

Class so far:

- + Model fitting
- + Model interpretation
- + Calculating predictions
- + Assessing model assumptions

What's next:

- + Assessing predictions
- + Variable selection
- + Inference

Warmup

Work on the warmup activity (handout), then we will discuss as a group

Class activity

Fitted model:

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = 3.78 - 0.037Age_i - 2.52Male_i - 1.31Class2_i - 2.58Class3_i$$

What is the predicted probability of survival for a male, first-class passenger aged 20? What about for a male, second-class passenger aged 30?

Making predictions with the Titanic data

- + For each passenger, we calculate $\hat{\pi}_i$ (estimated probability of survival)
- + But, we want to predict *which* passengers actually survive

How do we turn $\hat{\pi}_i$ into a binary prediction of survival / no survival?

Confusion matrix

```
m1 <- glm(Survived ~ Age + Sex + as.factor(Pclass),
           data = titanic, family = binomial)

table(Prediction = m1$fitted.values > 0.5,
       Truth = titanic$Survived)
```

```
##           Truth
## Prediction    0    1
##      FALSE 356   83
##      TRUE  68  207
```

Did we do a good job predicting survival?

Confusion matrix

```
##           Truth
## Prediction    0    1
##      FALSE 356   83
##      TRUE  68  207
```


Class activity

Predictions with the SBA data:

https://sta214-f25.github.io/class_activities/ca_10.html

Class activity

```
m1 <- glm(Default ~ log(DisbursementGross) + Term +  
          as.factor(UrbanRural),  
          family = binomial, data = sba)  
  
table(Prediction = m1$fitted.values > 0.5,  
      Truth = sba$Default)
```

```
##           Truth  
## Prediction FALSE TRUE  
##      FALSE  3989  734  
##      TRUE   100  168
```

- + Accuracy =
- + Sensitivity =
- + Specificity =
- + PPV =

Class activity

```
##              Truth
## Prediction FALSE TRUE
##      FALSE  3989  734
##      TRUE   100  168
```

Is an accuracy of around 80% good?

Class activity

Changing thresholds:

```
table(Prediction = m1$fitted.values > 0.3,  
      Truth = sba$Default)
```

```
##           Truth  
## Prediction FALSE TRUE  
##      FALSE  3524  351  
##      TRUE   565  551
```

```
table(Prediction = m1$fitted.values > 0.7,  
      Truth = sba$Default)
```

```
##           Truth  
## Prediction FALSE TRUE  
##      FALSE  4089  902
```